

APO-ATENIDONE TABLETS 100/25 MG

Atenolol 100mg and Chlorthalidone 25 mg Antihypertensive Agent

Product Description

White, round biconvex tablets. Engraved 100 over European bisect 25 on one side, and engraved APO on the other.

Pharmacology

APO-ATENIDONE (atenolol/chlorthalidone) combines the antihypertensive activity of two agents, a beta adrenergic receptor blocking agent (atenolol) and a diuretic (chlorthalidone).

Atenolol is a beta₁-selective, beta adrenergic blocking agent, devoid of membrane stabilizing or intrinsic sympathomimetic (partial agonist) activities. It is a racemic mixture and the beta₁ properties reside in the S(-) enantiomer. Beta₁-selectivity decreases with increasing dose.

The mechanism of the antihypertensive effect of atenolol has not been established. Among the factors that may be involved are:

- competitive ability to antagonize catecholamine-induced tachycardia at the beta receptor sites in the heart, thus decreasing cardiac output.
- Inhibition of renin release by the kidneys.
- Inhibition of the vasomotor centres.

Chlorthalidone, a monosulfonamyl diuretic, increases excretion of sodium and chloride. Natriuresis is accompanied by some loss of potassium. The mechanism by which chlorthalidone reduces blood pressure is not fully known but may be related to the excretion and redistribution of body sodium. Chlorthalidone usually does not decrease normal blood pressure.

The combination of atenolol with thiazide-like diuretics has been shown to be compatible and generally more effective than either drug used alone as an antihypertensive agent.

Pharmacokinetics: Approximately 40 to 50% of an oral dose of atenolol is absorbed from the gastrointestinal tract, the remainder being excreted unchanged in the feces. Peak plasma concentrations occur 2-4 hours after dosing and are subject to a 4-fold variability. The plasma levels are proportional to dose over the range 50-400 mg and 6 to 16% of atenolol is bound to plasma proteins. The plasma half-life is approximately 6-7 hours.

Approximately 60% of an oral dose of chlorthalidone is absorbed from the gastrointestinal tract and excreted unchanged in the urine following a single dose, the peak blood concentration of chlorthalidone occurs after approximately 12 hours and decreases thereafter according to first-order kinetics; the disposition half-life is approximately 50 hours. Approximately 75% of chlorthalidone is bound in plasma.

Indication

Management of hypertension.

Recommended Dose

Adults: 1 tab daily. Most patients with hypertension will give an adequate response to one tablet daily of APO-ATENIDONE. With increased dosage, there is little or no further fall in blood pressure, and if required, another antihypertensive drug, eg a vasodilator, can be added.

Children: Not recommended.

Elderly patients: Dosage requirements are frequently lower in this age group.

Renal failure: A decrease in daily dose or in frequency of administration may be needed in patients with severe renal impairment.

The effect of APO-ATENIDONE following a single oral dose is sustained for at least 24 hrs.

Contraindication

APO-ATENIDONE should not be used in patients with any of the following: Known hypersensitivity to either component; bradycardia; cardiogenic shock; hypotension; metabolic acidosis; severe peripheral arterial circulatory disturbances; second or third degree heart block; sick sinus syndrome; untreated phaeochromocytoma; uncontrolled heart failure.

Warnings and Precautions

Due to its beta-blocker component APO-ATENIDONE Tablets:

- although contraindicated in uncontrolled heart failure may be used in patients whose signs of heart failure have been controlled. Caution must be exercised in patients whose cardiac reserve is poor.

- may increase the number and duration of angina attacks in patients with Prinzmetal's angina due to unopposed alpha receptor mediated coronary artery vasoconstriction. Atenolol is a beta₁-selective beta-blocker; consequently the use of APO-ATENIDONE Tablets may be considered although utmost caution must be exercised.

- although contraindicated in severe peripheral arterial circulatory disturbances, may also aggravate less severe peripheral arterial circulatory disturbances.

- due to its negative effect on conduction time, caution must be exercised if it is given to patients with first-degree heart block.

- may modify the tachycardia of hypoglycaemia.

- may mask the signs of thyrotoxicosis.

- will reduce heart rate, as a result of its pharmacological action. In the rare instances when a treated patient develops symptoms which may be attributable to a slow heart rate, the dose may be reduced.

- should not be discontinued abruptly in patients suffering from ischaemic heart disease.

- may cause a more severe reaction to a variety of allergens, when given to patients with a history of anaphylactic reaction to such allergens. Such patients may be unresponsive to the usual doses of adrenaline used to treat the allergic reactions.

- may cause a hypersensitivity reaction including angioedema and urticaria.

APO-ATENIDONE Tablets contain the cardioselective beta-blocker atenolol. Although cardioselective (beta₁) beta-blockers may have less effect on lung function than non-selective beta-blockers, as with all beta-blockers, these should be avoided in patients with reversible obstructive airways disease, unless there are compelling clinical reasons for their use. Where such reasons exist, APO-ATENIDONE Tablets may be used with caution. Occasionally, some increase in airways resistance may occur in asthmatic patients, however, and this may usually be reversed by commonly used dosage of bronchodilators such as salbutamol or isoprenaline. The label and patient information leaflet for this product state the following warning:

“If you have ever had asthma or wheezing, you should not take this medicine

unless you have discussed these symptoms with the prescribing doctor”.

Due to its chlortalidone component:

- hypokalaemia may occur. Measurement of potassium levels is appropriate, especially in the older patient, those receiving digitalis preparations for cardiac failure, those taking an abnormal (low in potassium) diet or those suffering from gastrointestinal complaints. Hypokalaemia may predispose to arrhythmias in patients receiving digitalis;
- caution must be exercised in patients with severe renal failure;
- Impaired glucose tolerance may occur and caution must be exercised if chlortalidone is administered to patients with a known pre-disposition to diabetes mellitus;
- hyperuricaemia may occur. Only a minor increase in serum uric acid usually occurs but in cases of prolonged elevation, the concurrent use of a uricosuric agent will reverse the hyperuricaemia.

Drug Interaction

Combined use of beta-blockers and calcium channel blockers with negative inotropic effects e.g. verapamil, diltiazem, can lead to an exaggeration of these effects particularly in patients with impaired ventricular function and/or sino-atrial or atrio-ventricular conduction abnormalities. This may result in severe hypotension, bradycardia and cardiac failure. Neither the beta-blocker nor the calcium channel blocker should be administered intravenously within 48 hours of discontinuing the other.

Concomitant therapy with dihydropyridines e.g. nifedipine, may increase the risk of hypotension, and cardiac failure may occur in patients with latent cardiac insufficiency.

Digitalis glycosides, in association with beta-blockers, may increase atrio-ventricular conduction time.

Beta-blockers may exacerbate the rebound hypertension which can follow the withdrawal of clonidine. If the two drugs are co-administered, the beta-blocker should be withdrawn several days before discontinuing clonidine. If replacing clonidine by beta-blocker therapy, the introduction of beta-blockers

should be delayed for several days after clonidine administration has stopped.

Caution must be exercised when prescribing beta-blockers with Class 1 antiarrhythmic agents such as disopyramide.

Concomitant use of sympathomimetic agents, e.g. adrenaline, may counteract the effect of beta-blockers.

Concomitant use with insulin and oral antidiabetic drugs may lead to the intensification of the blood sugar lowering effects of these drugs.

Concomitant use of prostaglandin synthetase inhibiting drugs (e.g. ibuprofen, indomethacin) may decrease the hypotensive effects of beta-blockers

Preparations containing lithium should not be given with diuretics because they may reduce its renal clearance.

Caution must be exercised when using anaesthetic agents with APO-ATENIDONE Tablets. The anaesthetist should be informed and the choice of anaesthetic should be an agent with as little negative inotropic activity as possible. Use of beta-blockers with anaesthetic drugs may result in attenuation of the reflex tachycardia and increase the risk of hypotension. Anaesthetic agents causing myocardial depression are best avoided.

Use in pregnancy and lactation

Apo-Atenidone should not be given during pregnancy and lactation.

Adverse Effects

APO-ATENIDONE Tablets are well tolerated. In clinical studies, the undesired events reported are usually attributable to the pharmacological actions of its components.

The following undesired events, listed by body system, have been reported with APO-ATENIDONE Tablets or either of its components:

Biochemical:

hyperuricaemia, hypokalaemia, impaired glucose tolerance, hyponatraemia related to chlortalidone.

Cardiovascular:

Bradycardia, heart failure deterioration, postural hypotension which may be associated with syncope, cold extremities. In susceptible patients: precipitation of heart block, intermittent claudication, Raynaud's phenomenon.

CNS:

Confusion, dizziness, headache, mood changes, nightmares, psychoses and hallucinations, sleep disturbances of the type noted with beta-blockers

Gastrointestinal:

dry mouth, gastrointestinal disturbances, nausea. Elevations of transaminase levels have been seen infrequently, rare cases of hepatic toxicity including intrahepatic cholestasis have been reported. Pancreatitis.

Haematological:

Leucopenia, purpura, thrombocytopenia.

Integumentary:

Alopecia, dry eyes, psoriasiform skin reactions, exacerbation of psoriasis, skin rashes.

Neurological:

paraesthesia.

Respiratory:

bronchospasm may occur in patients with bronchial asthma or a history of asthmatic complaints.

Reproductive:

impotence

Special senses:

visual disturbances.

Others:

hypersensitivity reactions, including angioedema and urticaria; fatigue, an increase in ANA (Antinuclear Antibodies) has been observed, however the clinical relevance of this is not clear.

Discontinuance of APO-ATENIDONE Tablets should be considered if, according to clinical judgement, the well-being of the patient is adversely affected by any of the above reactions.

Symptoms and Treatment of Overdose

The symptoms of overdosage may include bradycardia, hypotension, acute cardiac insufficiency and bronchospasm.

General treatment should include: close supervision, treatment in an intensive care ward, the use of gastric lavage, activated charcoal and a laxative to prevent absorption of any drug still present in the gastrointestinal tract, the use of plasma or plasma substitutes to treat hypotension and shock. The possible use of haemodialysis or haemoperfusion may be considered.

Excessive bradycardia can be counteracted with atropine 1-2 mg IV. This may be followed by a bolus

dose of glucagon 10 mg IV if needed. This may be repeated or followed by an IV infusion of glucagon 1-10 mg/hr depending on reaction if necessary. If no response to glucagon occurs or if glucagon is unavailable, a β -adrenoceptor stimulant, eg. dobutamine 2.5 – 10 μ g given as an infusion at a rate not exceeding 5 μ g/min may be given. It is likely that these doses would be inadequate to reverse the cardiac effects of β -blockade if a large overdose has been taken. The dose of dobutamine or isoprenaline should therefore be augmented if essential to achieve the required reaction in accordance with the clinical condition of the patient.

There is a probability of hypotension will be reduced by the use of the more selective agents, prenalterol and dobutamine.

Excessive diuresis should be counteracted by preserving normal fluid and electrolyte balance.

Storage Conditions

Store below 30°C.

Availability of Dosage Forms

Apo-Atenidone 100/25 tablets: white, round, biconvex tablets. Scored and engraved "100" over "25" on one side and engraved "APO" on the other. Available in blisters of 30 tablets.

Manufacturer

Apotex Inc., 150 Signet Dr, Toronto (Weston), Ontario, Canada M9L 1T9.

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